Lab 3
Localization using Unscented Kalman Filter

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1. Description of the UKF

The Unscented Kalman Filter (UKF) maintains the mean of the belief over robot pose, $\mu = [x,y,\theta]$, and a 3x3 covariance matrix, Σ. The UKF alternates between a prediction step, which is the result of wheel odometry, and an update step which is the result of LIDAR scans of beacons. The UKF's main contribution is how it propagates uncertainty through nonlinear functions. The UKF, as opposed to the EKF, does not rely on the use of the Jacobians to linearize the function and the motion model. The UKF relies on the Unscented Transform. This uses a total of 7 points, which is $2n + 1$, that are centered around the mean, passes them through the nonlinear function, and then reconstructs a Gaussian using the points with a mean and outer product of the weights. The UKF, unlike the EKF, does not require derivatives and takes into account nonlinear effects during the 3rd order as opposed to the 1st order as done in the EKF.

Compared with the EKF from Lab 2, Jacobians were neither computed nor coded in this case. The same motion model $f(x, u)$ and measurement model $h(x)$ from Lab 2 are reused completely unchanged, and are called as black-box functions for all 7 sigma points. During the predicted step, sigma points are taken from the current belief and then pushed through the unicycle motion model, and recombined to construct the predicted mean μ' and covariance Σ' (plus process noise Q). During the update step, residual sigma points are taken from the predicted mean, pushed through the range-and-bearing measurement model for the corresponding matched beacon, and then used to determine the predicted measurement $\hat{z}$, the innovation covariance S and the cross-covariance T. The Kalman gain $K = T S^{-1}$ is then used to modify the mean and reduce the covariance. Angle wrapping with wrap_to_pi() is used at each step where a bearing difference is calculated to avoid divergences at $\theta = \pm\pi$.

2. Simulation Results

2.1 Baseline Run (Clean Odometry)

The circle_driver script was used to move the robot in a circular path within the Gazebo simulation environment. The Unscented Kalman Filter (UKF) was provided with undisturbed odometry data from the /odom topic and LIDAR detections for three known beacons located at positions (3,0), (0,3), and (-3,1.5). Of the three UKF estimates, the one given the LIDAR data showed the best tracking performance. The blue and green lines became almost completely overlapping. The odometry drifted noticeably inwards along the trajectory and was represented as a red dashed line.

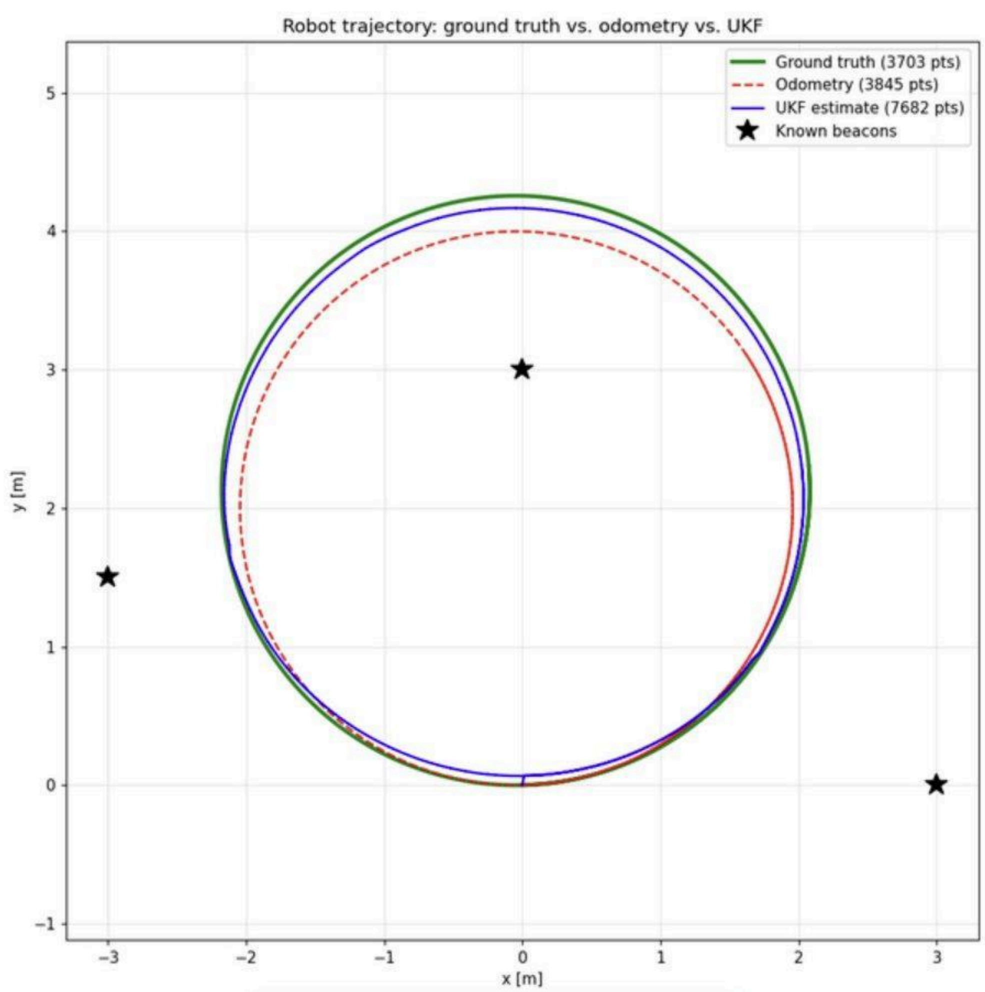

Figure 1: Ground truth (green), odometry (red dashed), UKF estimate (blue): clean odometry run.

2.2 Noisy Odometry Experiment

The UKF was switched to /noisy_odom, which is a stream of odometry that has Gaussian noise orders of magnitude larger than those typically used, courtesy of the noisy_odom_publisher node. In the 4 line plot below, you can see the noisy odometry (orange dotted) underwent a dramatic drift and spiraled outward, while the UKF (blue) maintained a true circular path and fused LIDAR beacon measurements.

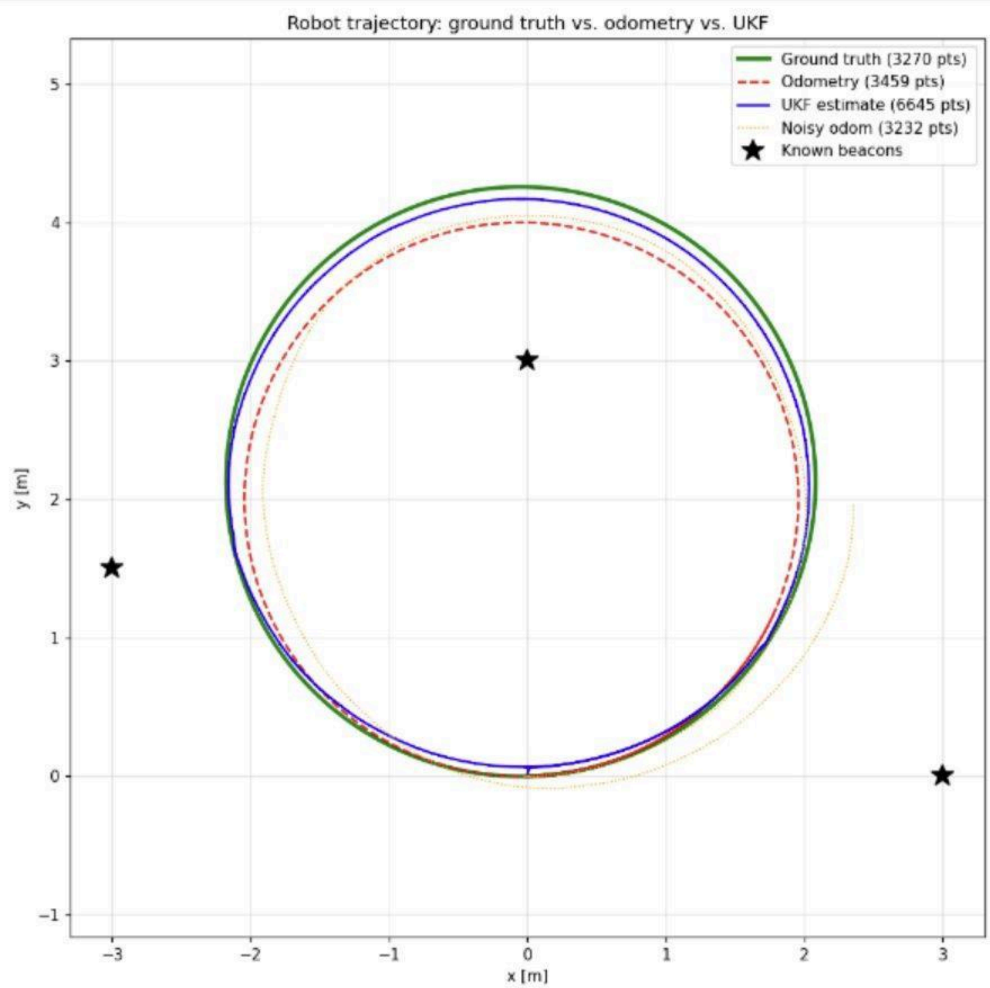

Figure 2: Ground truth (green), clean odometry (red dashed), noisy odometry (orange dotted), UKF on noisy input (blue).

2.3 Bonus: UKF vs. EKF Side-by-Side

Both the UKF and Lab 2's EKF ran videos at the same time, both subscribing to /noisy_odom. The 5-line plot shows that the UKF (blue) tracked the ground truth better than the EKF (purple) given the same conditions for noisy odometry. The following Gazebo screenshot has the robot with 3 beacons mid-trajectory.

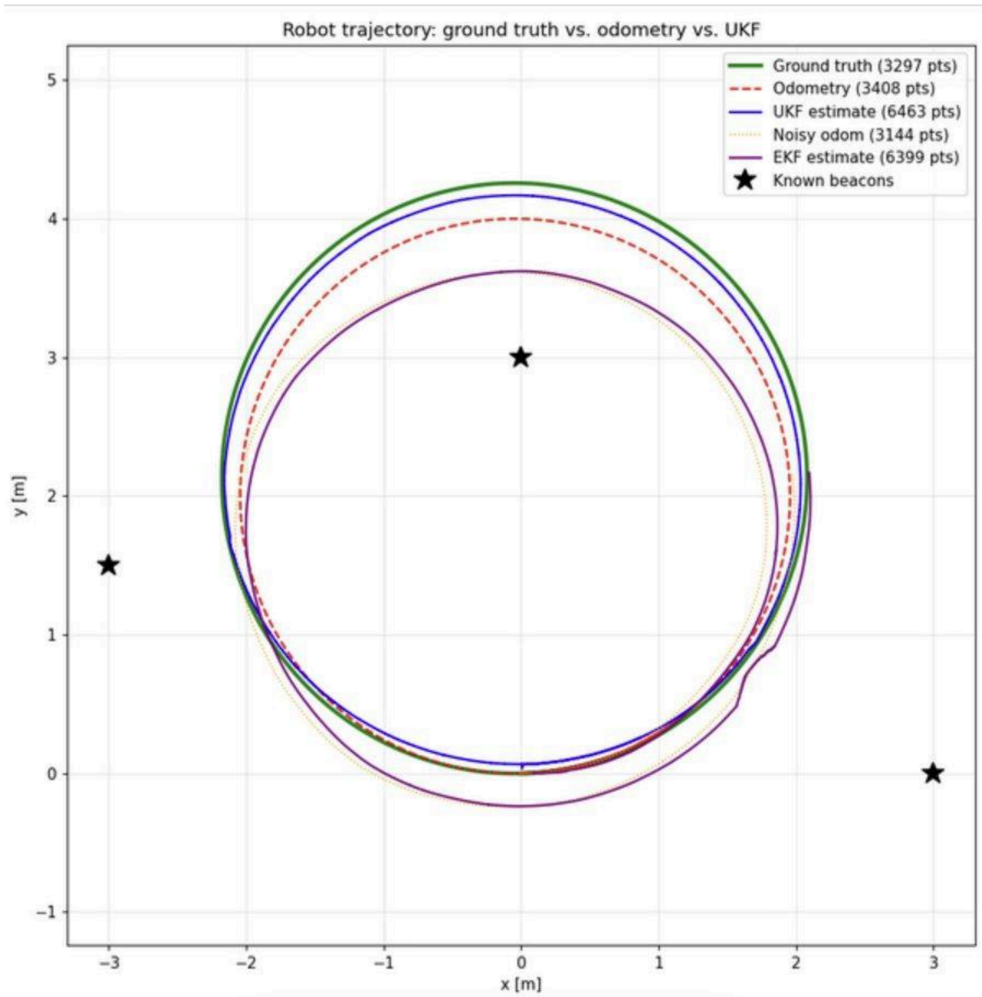

Figure 3: All five traces: truth (green), clean odom (red), noisy odom (orange), UKF (blue), EKF (purple).

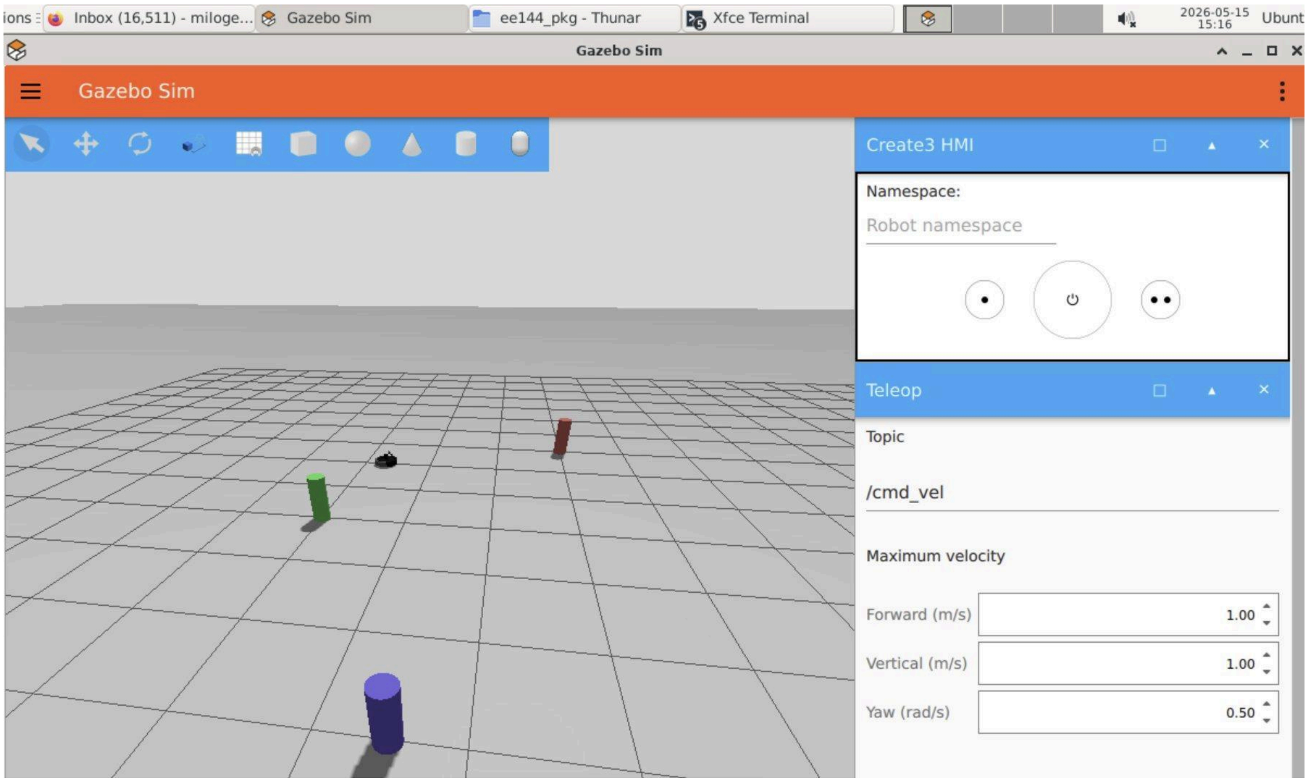

Figure 4: Gazebo simulation showing the Turtlebot4 and three known beacons mid-trajectory.

3. Discussion

Throughout all experiments, the UKF was able to track ground truth more satisfactorily than odometry. In the clean odometry instance, the blue UKF estimate was almost fully overlapped by the green ground-truth line, and the red odometry line drifted inwards by a visible line over the course of the circular trajectory. This validates that the fusion of LIDAR beacon measurements through the UKF update step is an efficient and effective way to correct accumulated odometric drift.

The UKF was still able to produce reasonable estimates that were still in close proximity to ground truth, even with the addition of naturally drifting and noisy odometry. Although the orange noisy odometry drifted severely and began to spiral outwards, the blue UKF line had retained its circular shape and scale. This provides evidence that the filter is able to down-weight odometry predictions and is more reliant on LIDAR beacon measurements to adjust the irregular odometry.

The UKF's performance under naturally noisy odometry was superior to the EKF's performance in this regard. The EKF was forced to assume that the odometry predictions were correct within a small enough region from the prediction. Under high velocity noise, such an assumption of small noise is impossible. The UKF does not have this problem, as it recalibrates using sigma points, regardless of the size of the odometry predictions. This is due to the UKF's assumption of controlling the boundaries of the motion model.

The UKF's covariance Σ behaved in the UKF as it did in the EKF. It grew in the predicted step and diminished with each measurement. Noisy odometry caused Σ to grow more during predicted steps. High velocity uncertainty led to more process noise Q, meaning the noise began to predominate. With the UKF, sigma-point propagation was better, and even with noisy odometry the covariance remained well-posed. With the EKF, covariance tended to underestimate uncertainty from bias which lead to worse position calculation.